

# BAAZ & UDAY JKBOSE

2026 CLASS 11<sup>TH</sup>

## BIOMOLECULES

LECTURE-1

ZOOLOGY

DAVOOD SIR (CEO of NG)



# Topics To Be Covered

- 1 How to Analyse Chemical Composition?
- 2 Metabolites (Primary and Secondary Metabolites)

# What are Biomolecules?

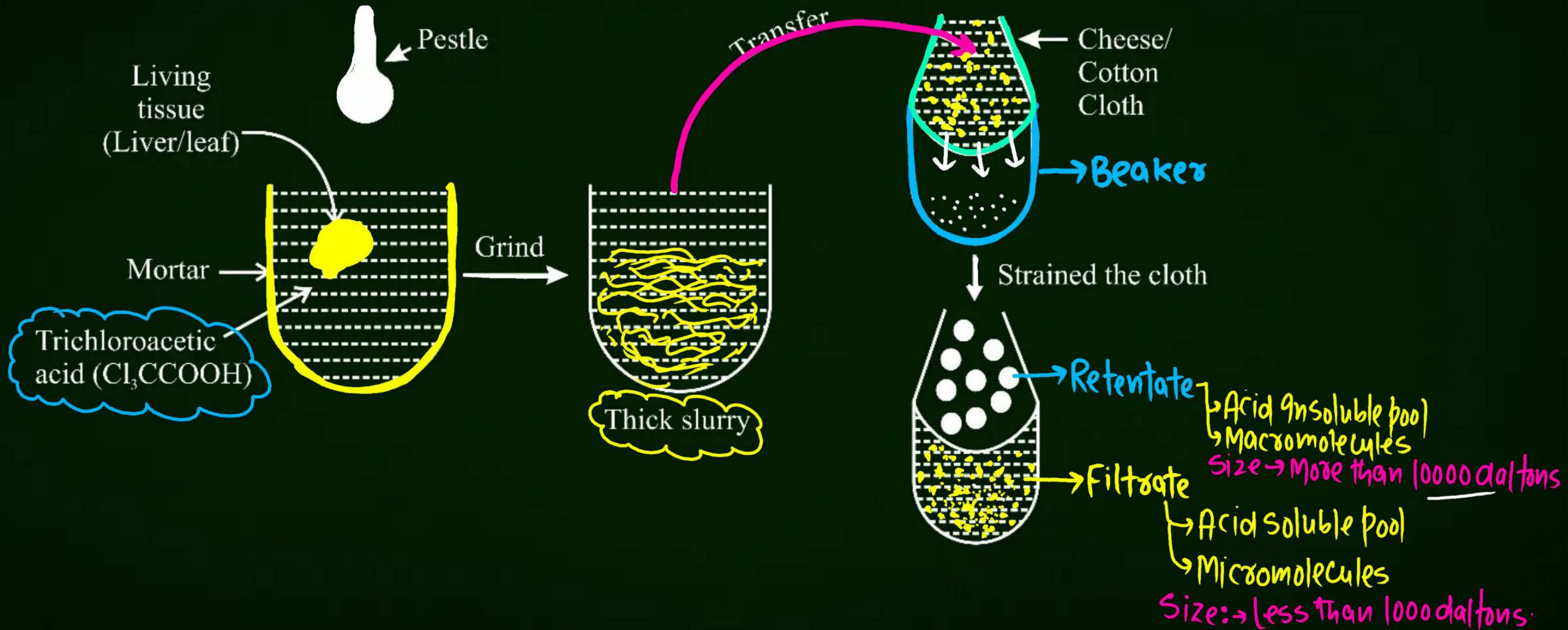
Living + Chemical Substance

- All the Carbon Containing Compounds present in a living organism are called biomolecules.

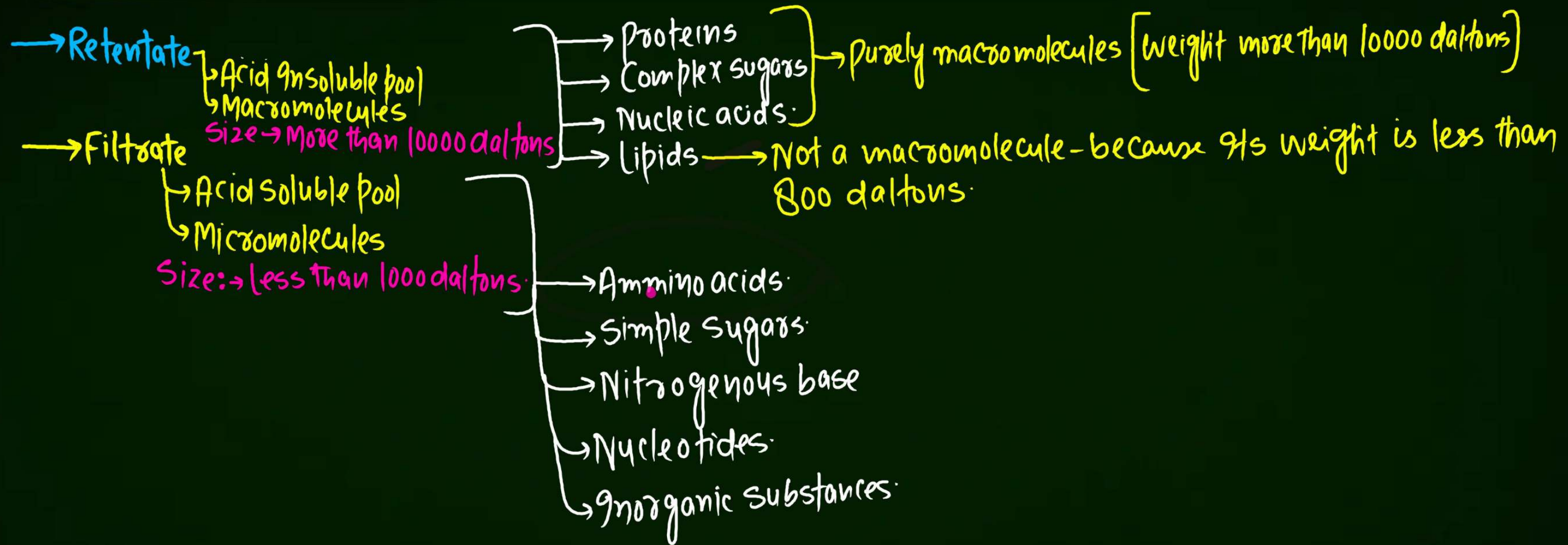
# How to Analyse Chemical Compounds?

|                                     |                      |                                                                |
|-------------------------------------|----------------------|----------------------------------------------------------------|
| ➤ Animal tissues                    | Chemical<br>Analysis | Similar types of chemicals are present But in different amount |
| ➤ Plant tissues                     |                      |                                                                |
| ➤ Micro-organisms                   |                      |                                                                |
| ➤ Human tissue<br>(Living matter)   | Chemical<br>Analysis | Similar types of chemical are present But in different amount  |
| ➤ Earth crust<br>(Nonliving matter) |                      |                                                                |

# How to Analyse Chemical Compounds?



# How to Analyse Chemical Compounds?



# A comparison of Elements present in Non livings & Livings

| Element        | % Weight of Earth's crust Human body |                            |
|----------------|--------------------------------------|----------------------------|
| Hydrogen (H)   | 0.14                                 | 0.5 <i>Text book error</i> |
| Carbon (C) ✓   | 0.03                                 | 18.5                       |
| Oxygen (O)     | 46.6                                 | 65.0                       |
| Nitrogen (N)   | Very little                          | 3.3                        |
| Sulphur (S)    | 0.03                                 | 0.3                        |
| Sodium (Na)    | 2.8                                  | 0.2                        |
| Calcium (Ca)   | 3.6                                  | 1.5                        |
| Magnesium (Mg) | 2.1                                  | 0.1                        |
| Silicon (Si)   | 27.7                                 | Negligible                 |

**Does Livings have only organic compounds?**

→ Ans: No



## ASH ANALYSIS TEST

Green Leaf

↓ weight

We obtain wet weight.

↓ Heat

We obtain dry weight.

↓ Burn

organic part is burnt

Remaining ash containing  
inorganic compounds like  
Calcium, Magnesium etc

# Metabolites

The substances used or formed during metabolic Rxn in an organism are known as metabolites

## Metabolites

### Primary Metabolite

Metabolites formed in organism which has a direct influence or control on the physiological process

E.g Glucose, Ammino acid  
Chlorophyll.

### Secondary metabolite

Metabolites formed and found in certain plants, fungi, microbes. Which doesn't have a direct control on their physiological process but can be beneficial for humans

E.g Carotenoids.  
Alkaloids.  
Plant oils.  
• Toxins.

**Thank**  
*You*